FINESST Concept Brief

Expanding Barlow (2026) into a Future Investigator Research Project

NASA ROSES-2025 F.5 — Future Investigators in NASA Earth and Space Science and Technology | Proposal deadline: 14 July 2026

1. The parent work

Barlow (2026) trained a U-Net to delineate ephemeral-stream boundaries in the McMurdo Dry Valleys (MDVs), Antarctica, **from terrain shape alone** (elevation, slope, aspect — no water/color signal), validated free satellite elevation models (REMA) against airborne lidar for centimeter-to-meter change detection, and quantified two decades of fluvial geomorphic change across four valleys. The dissertation rests on three transferable pillars:

- **Terrain-only semantic segmentation** of channels from DEM derivatives (elevation/slope/aspect most informative).
- **Satellite-DEM validation** via point-to-plane ICP alignment, Laplacian error modeling, and NMAD-based (outlier-robust) uncertainty.
- **Multi-epoch change detection** — DEM-of-Difference inside segmented stream masks, level-of-detection thresholds (LOD95 = 1.96 × NMAD), and per-stream erosion/deposition and acceleration rates.

Citation: Barlow, M. C. (2026). *A Comprehensive Spatial Analysis of Stream Boundary and Geomorphological Change Detection: McMurdo Dry Valleys, Antarctica.* PhD Dissertation, Dept. of Civil & Environmental Engineering, University of Houston.

2. The opening (core move)

Barlow's thesis ends at *description* — it maps where change is happening (Denton Hills is the erosion hotspot; parts of Taylor Valley are accelerating) but explicitly defers two threads to future work: **(a)** coupling geomorphic change to physical / climate drivers, and **(b)** generalizing the model beyond the MDVs. A Future Investigator (FI) project lives precisely in that gap: turn her *monitor* into an *explanatory and predictive* tool. This is a genuine increment, not a replication — exactly what FINESST rewards.

3. Proposed FI project

Target division: Earth Science (EARTH25) — continuing the Antarctic cryosphere / climate thread.

“From detection to attribution: physically-coupled, cross-domain monitoring of ephemeral-channel geomorphic change from terrain alone.”

#	Objective	Increment over Barlow

1	Attribution (headline science)	Pair per-stream gross / net / acceleration rates with energy-balance drivers (temperature, insolation, melt degree-days, active-layer depth from MDV-LTER + reanalysis); model why hotspots occur and predict where acceleration migrates next. Listed verbatim as Barlow future work.
2	Cross-sensor & cross-valley generalization	Make the terrain-only segmenter robust across sensors (airborne lidar 2001 / 2014 and REMA satellite) and across all four valleys — quantifying and correcting the lidar-to-satellite domain shift that limits the satellite epoch. Barlow flags sensor generalization as future work.
3	Calibrated uncertainty	Extend the Laplacian / NMAD / LOD framework into a per-pixel, sensor-aware confidence layer so every change map carries calibrated uncertainty — aligned with NASA's trustworthy-EO-product priorities.

4. Why it is competitive (maps to the review criteria)

- **Scientific merit.** Advances the MDV “climate canary” from *measured* to *explained and predicted*; novel attribution plus calibrated-uncertainty change detection.
- **Relevance to SMD.** Quantifying cryosphere and climate sensitivity in Earth's most geomorphically stable landscape maps squarely onto NASA Earth Science Division objectives (cryosphere, terrestrial hydrology, climate change).
- **FI qualifications & development.** The applicant already operates the complete toolchain — U-Net segmentation on terrain derivatives, ICP alignment, DEM-of-Difference — in an independent application domain (see Feasibility), a concrete anchor most FI applicants lack.

5. Feasibility anchor

An existing, working pipeline (“WellSight”) already runs the identical architecture — U-Net semantic segmentation on lidar-derived terrain rasters (DEM, slope, local relief, openness), ICP-based co-registration, and DEM-of-Difference change analysis — on a completely different landscape (Appalachian Plateau channels, roads, and well-pad scars detected from elevation alone). This demonstrates the method generalizes beyond a single sensor and biome, directly de-risking Objective 2 and evidencing the FI's readiness to execute.

6. Data requirements

The pipeline runs on DEM-derived rasters, so elevation data is the backbone. Nearly all of it is free / public; the single access-gated dependency is Barlow's hand-digitized training labels. None is currently held locally — the existing WellSight holdings serve only as method-feasibility evidence.

A. Core terrain / elevation inputs

- **MDV airborne lidar, 2001 & 2014** (NCALM; via OpenTopography) — Barlow's gold-standard epochs.

- **REMA** (Reference Elevation Model of Antarctica), 2021–2023 — Polar Geospatial Center, free — the satellite epoch.
- Derived per epoch: elevation, slope, aspect, curvature, intensity (lidar), flow accumulation.

B. Training labels (ground truth)

- **Barlow's hand-digitized stream-boundary polygons** (217 Taylor tiles + the multi-valley set) — **the only access-gated dependency**: obtain from her group, or re-digitize.
- MCM-LTER stream centerlines as auxiliary / QC.

C. Climate & physical-driver data — required for the new science (Objective 1, attribution)

- **MDV-LTER meteorological network** — air/ground temperature, incoming solar radiation, humidity, wind (free, MCM-LTER data portal).
- LTER stream-discharge gauges (melt-season flow); glacier mass-balance / melt; permafrost / active-layer measurements.
- Reanalysis for spatial gap-fill: **ERA5** and/or **AMPS** (Antarctic Mesoscale Prediction System).

D. Cross-sensor / cross-valley data (Objective 2)

- All four valleys (Taylor, Wright, Victoria / Barwick, Denton Hills) across both lidar epochs and the REMA epoch — the spread needed to test generalization within the MDV system.
- The lidar-vs-REMA pairing is itself the key test case: quantify and correct the airborne-to-satellite domain shift that currently limits the satellite epoch.
- Optional broader reach (still Antarctic / polar): other polar-desert DEMs via ArcticDEM (PGC), if a wider generalization test is wanted.

E. Validation / auxiliary

- High-resolution imagery for label QC — WorldView / Maxar via PGC over the MDVs.
- Published geomorphic-change rates for cross-validation.

Status: none held locally yet, but almost all is free / public (REMA & ArcticDEM from PGC, MDV-LTER climate, airborne lidar from OpenTopography). The lone true dependency is Barlow's training labels.

7. Program logistics

- **Solicitation:** NASA ROSES-2025 F.5 (FINESST) — **Earth Science Division (EARTH25)**. (FINESST spans five SMD divisions; this proposal targets Earth Science.)
- **Deadline:** 14 July 2026 (11:59 PM EDT). Pre-proposal webinar 28 May 2026; Earth Science office hours 23–24 June 2026.
- **Structure:** the graduate student is the Future Investigator, primary author, and intellectual lead; advisor serves as PI of record. Award up to ~$50,000/yr for up to three years.

- **Deliverable to draft next:** ~6-page Scientific/Technical/Management section (Background & gap → objectives/hypotheses → methodology → expected results & novelty → FI role & development → timeline), references, and data-management plan.

8. Open decisions before drafting the full proposal

- **Data access:** whether Barlow's MDV training labels and the lidar / REMA DEMs can be used and extended directly, or the proposal must assume re-deriving them.
- **Scope:** which valleys and epoch pairs to prioritize for the attribution analysis, given REMA's uneven post-2014 coverage.

Prepared as a planning brief; not a submitted proposal. Sources: Barlow (2026) dissertation; NASA ROSES-2025 F.5 FINESST solicitation (NSPIRES).